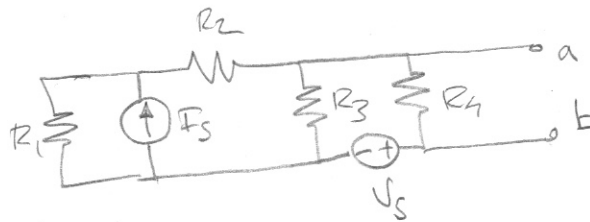


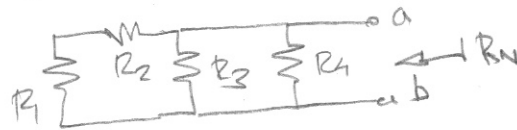
3.69)



$$\begin{aligned} R_1 &= 6 \, \Omega \\ R_2 &= 3 \, \Omega \\ R_3 &= 2 \, \Omega \\ R_4 &= 2 \, \Omega \\ V_s &= 10 \, \text{V} \\ I_s &= 3 \, \text{A} \end{aligned}$$

Bestäm en Nortonekvivalent
till nätverket ovan.

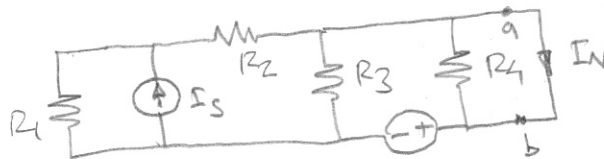
Lösning: Först $R_{TH} = R_N$



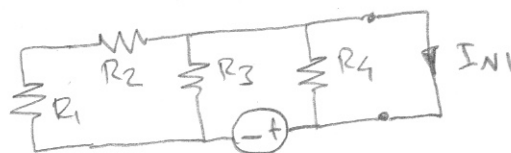
$$R_N^{-1} = \frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2 + R_4} = \frac{1}{2} + \frac{1}{2} + \frac{1}{9} = \frac{10}{9}$$

$$R_N = \frac{9}{10} \, \Omega$$

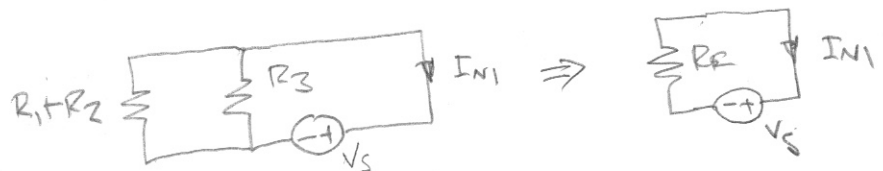
Sedan tar vi kortslutningsströmmen I_N



Superpos principen V_s verkar bra att använda
första halvan



R_4 är kortslut V_s så det tar vi bort

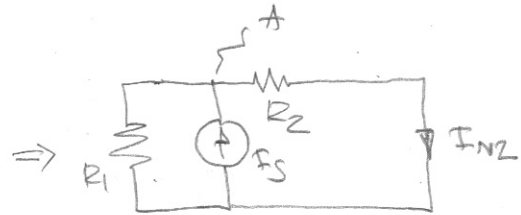
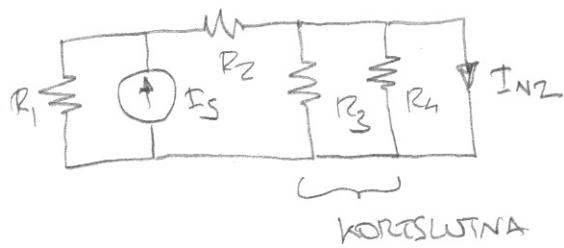


$$R_E = \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3} = \frac{(6+3)2}{6+3+2} = \frac{18}{11} \, \Omega$$

$$\text{KSLP ger } -V_s - R_E I_{N1} = 0$$

$$\begin{aligned} I_{N1} &= -\frac{V_s}{R_E} = -\frac{10}{18} \, \text{A} \\ &= \frac{10}{18} \approx \underline{\underline{0.11 \, \text{A}}} \end{aligned}$$

ANDRA HALVAN



SPÄNNINGEN I NODEN A.

TES AV

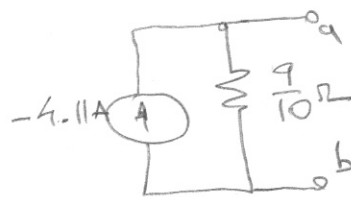
$$V_A = \frac{R_1 R_2}{R_1 + R_2} I_S =$$

$$= \frac{6 \cdot 3}{6 + 3} = 6 \text{ VOLT}$$

$$\Rightarrow I_{N2} = \frac{V_A}{R_2} = \frac{6}{3} = \underline{\underline{2 \text{ A}}}$$

SAMMANLÄGGER FÖR STRÖMMEN

$$I_N = I_{N1} + I_{N2} = -6.11 + 2 = \underline{\underline{-4.11 \text{ A}}}$$



$$\underline{\underline{SVAR:}} \begin{cases} R_N = \frac{9}{10} \Omega \\ I_N = -4.11 \text{ A} \end{cases}$$